

Auckland Road Crash

Analytics

MBI908 Capstone Project
Data Analytics & Implementation

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Project at a Glance

111,657

Crash Records

2015–2025

Data Period

10 Years

Historical Span

9

Notebooks

Fabric Pipeline

9 Gold
Tables

Delta Lake Output

12 Recs

Business Actions

Presentation Overview

01		Industry Problem & Context Why Auckland crash data matters and what gap this project fills	Slide 3
02		Domain Knowledge & Limitations Safe System, CRISP-DM, Nielsen — what worked and what didn't	Slide 4
03		Research Methodology How the data was collected, cleaned, and analysed	Slide 5
04		Key Findings 4 analytical streams: geospatial, ML, temporal, vulnerable users	Slide 6
05		Solution & Recommendations 12 evidence-based actions with NZD ROI estimates	Slide 7
06		Ethical & Cultural Safeguards Data governance, Te Tiriti, bias, mitigation strategies	Slide 8
07		Conclusion & Next Steps Contributions, limitations, future research	Slide 9

01 Industry Problem & Context

The Problem

No systematic use of CAS data for spatial + ML analysis in Auckland



Auckland Transport lacks integrated crash intelligence tools

NZD 4B+ annual social cost

Each fatality = NZD 4.84M; each serious injury = NZD 662K (NZTA, 2024)

Road to Zero: -40% deaths/injuries by 2030

Evidence-based intervention prioritisation is essential

111,657 Auckland crashes (2015-2025)

4.8% result in fatal or serious injury — but WHERE and WHY?

Vulnerable users at 8.95× higher risk

Pedestrians, cyclists, motorcyclists severely underprotected

Project Scope

Dataset: NZTA CAS WGS84 2015-2025

Platform: Microsoft Fabric (Medallion Architecture)

Tools: Python, scikit-learn, XGBoost, SHAP, folium, Power BI

Stakeholders: NZTA, Auckland Transport

Research Questions

RQ1 Where are Auckland's statistically significant crash hotspots?

RQ2 Can ML reliably predict crash severity from road & environment data?

RQ3 Do holiday periods significantly affect crash severity?

RQ4 What factors most predict severe outcomes for vulnerable users?

02 Domain Knowledge - What Applied, What Didn't, and Why

LO1 - Critical evaluation of domain-specific knowledge

Safe System Framework (NZTA, 2020)



✓ **Applied:** Organised analysis around 4 pillars: safe roads, safe speeds, safe vehicles, safe road use. Ensured systemic rather than individual-blame framing.

⚠️ **Limits:** Public CAS excludes driver behaviour data (alcohol, speed). 'Safe vehicles' pillar unanalysable. SHAP features used as proxy for road environment vs driver context contribution.

CRISP-DM Process Model



✓ **Applied:** Medallion architecture (Bronze→Silver→Gold) maps naturally to CRISP-DM phases. Structured iterative development across 9 analysis notebooks.

⚠️ **Limits:** No formal Deployment phase implemented for ML model. high_risk_flag available in Gold layer but no production API or Police integration built.

Nielsen Usability Heuristics (Nielsen, 1994)

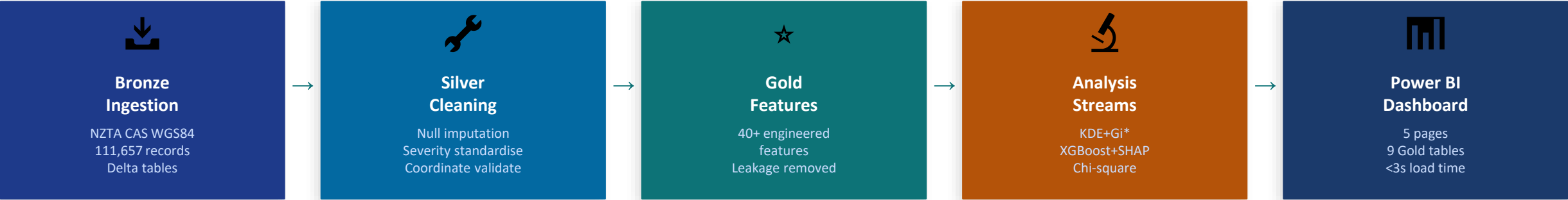


✓ **Applied:** All 10 heuristics applied in Power BI design. Plain language labels (not database codes), consistent severity colours, mandatory data limitation note cards.

⚠️ **Limits:** Standard Power BI templates don't include methodology transparency features. Required explicit design overrides for heuristics 2 and 10.

03 Research Methodology -Justified Against Real-World Constraints

LO2 - Methodological justification against feasibility, access, cost, time, and ethics



Research Constraints & How They Were Managed

Constraint	Detail	How Managed	Impact on Quality
No crash hour data	Public CAS excludes time-of-day	Temporal scope narrowed to holiday periods; note card in dashboard	Temporal findings limited to annual + holiday comparisons
No driver behaviour data	No speed, alcohol, or licence data	ML ceiling at ~0.72 AUC documented; SHAP confirms road features meaningful	AUC target revised; framed as data-constrained not model-constrained
No Fabric Premium	Cannot use Fabric workspace connections	9 Gold tables exported to CSV for Power BI Desktop import	Full functionality preserved; minor workflow step added
Class imbalance (4.8% severe)	Severe crashes rare in training data	XGBoost scale_pos_weight; threshold=0.35 to maximise recall	Precision lower but recall 63.8% — appropriate for safety use case

04a Finding 1 - Geospatial Hotspot Analysis (RQ1)

36 statistically significant hotspot grid cells — SH1/Southern Motorway Corridor confirmed at 99.9% confidence ($G_i^* Z > 3.29, p < 0.001$)

Hot Spot 99.9%	Hot Spot 99%	Hot Spot 95%	Cold Spot 95%
 12 cells Grafton–CBD Eden Terrace	 9 cells SH1 Southern Motorway	 15 cells Henderson New Lynn	 7 cells Northern suburbs

Top 5 KDE Hotspot Locations					Key Insight KDE and G_i^* independently confirm the same zones - cross-method validation strengthens confidence. Moran's $I > 0, p < 0.05$ confirms crashes cluster spatially, not randomly.
#1	Karangahape Road	Eden Terrace	1,497	Hot Spot 99.9%	
#2	Mt Wellington Highway	Mount Wellington	749	Hot Spot 99.9%	
#3	Great South Road	Clover Park	573	Hot Spot 99%	
#4	Weymouth Road	Manurewa	544	Hot Spot 99%	
#5	Northern Motorway	Forrest Hill	355	Hot Spot 95%	

04b Finding 2 - Machine Learning Severity Prediction (RQ2)

Model Comparison Results			
Model	ROC-AUC	Recall	Rationale
Logistic Regression	0.731	0.582	Highest AUC - baseline
Random Forest	0.712	0.427	Lower recall

⚠️ Why is AUC ~0.72 and not 0.85? Public CAS lacks crash hour, driver speed, and alcohol data the key severity predictors identified in literature. This is a **data ceiling, not a model failure**. All three models converge at ~0.71-0.73, confirming the dataset has been exhausted from a feature perspective. XGBoost is selected for its superior recall (63.8%) the operationally critical metric for safety-focused use.

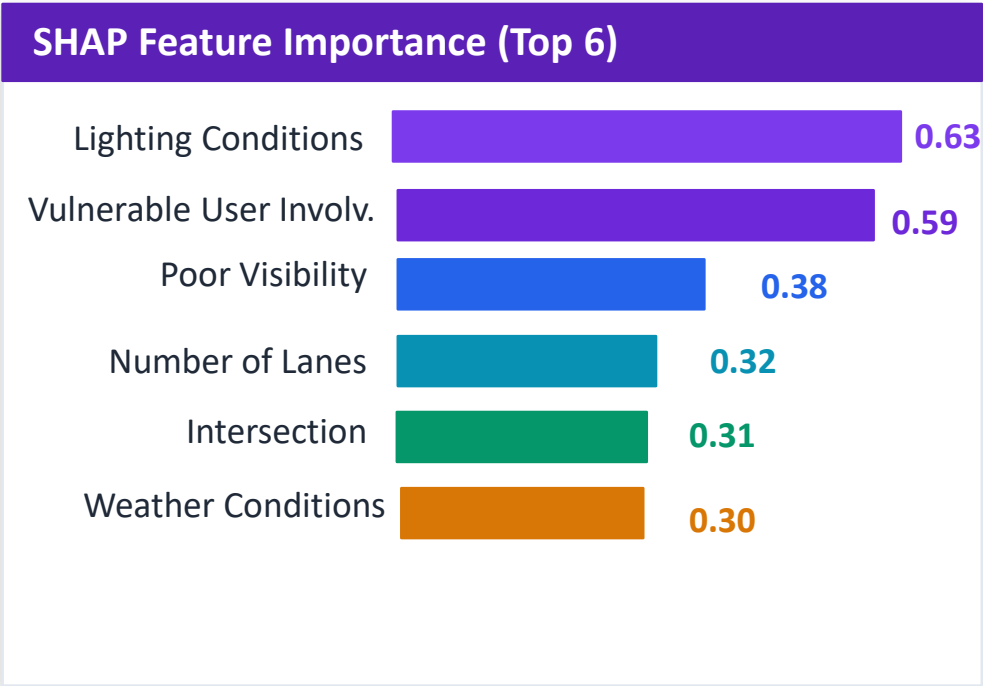
Deployment Decision: Why XGBoost?

📌 Recall = 63.8%

Identifies 6 in 10 actual severe crashes - critical for safety patrol routing

🔍 high_risk_flag applied at threshold = 0.35

Flags 38% of crashes for enhanced attention with manageable false positive rate



5-Fold Cross-Validation

CV AUC: 0.718 ± 0.009

CV Recall: 0.635 ± 0.012

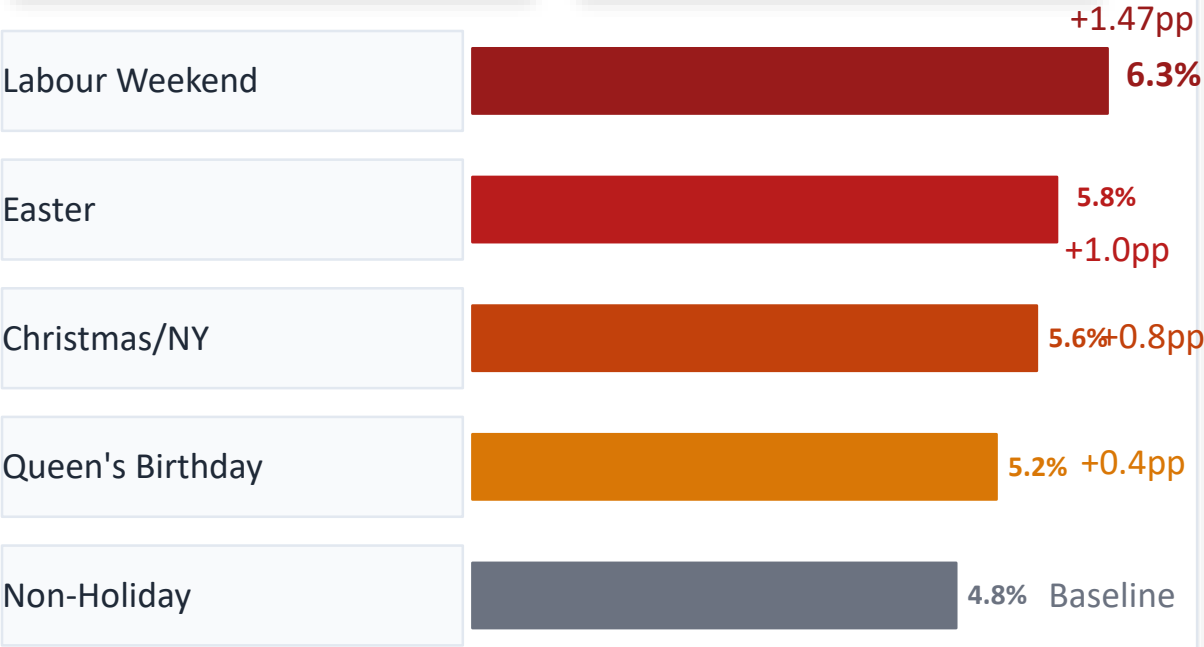
Low std confirms: no overfitting - results generalise

04c Findings 3 & 4 - Temporal Patterns & Vulnerable User Risk

RQ3 - Holiday Period & Temporal Patterns

$\chi^2=10.09$
p=0.018 ✓

$V=0.0095$
Negligible effect



⚠️ **Note:** Crash hour is NOT available in the public NZTA CAS dataset. Temporal analysis is limited to annual trends and holiday-period comparisons. Time-of-day patterns cannot be assessed from this data source alone.

RQ4 - Vulnerable User Risk (8.95× Relative Risk)

8.95×

More likely to result in fatal or serious injury

VU severe rate: 23.7% vs Non-VU: 2.6%

Pedestrian 4,598	Highest CBD concentration
Motorcyclist 4,573	Widest geographic spread
Cyclist 2,510	CBD commuter + recreation

Top Risk Factors (Cramér's V)



05 Solution - Integrated Road Safety Intelligence Platform



Component 1 Fabric Analytics Pipeline

Bronze→Silver→Gold medallion
9 Gold Delta tables
111,657 records processed
GitHub version control



Component 2 Power BI Dashboard

5-page stakeholder dashboard
Geospatial · ML · Temporal
Vulnerable Users · Executive
<3s load time



Component 3 12-Recommendation Plan

Evidence-based actions
Each linked to findings
NZD ROI estimates
Priority matrix

Combined Annual ROI Estimate · 3 Independent Intervention Streams

~NZD 8M

Stream 1: Hotspot Infrastructure
(5% crash reduction at Gi* hot spot sites)

~NZD 7M

Stream 2: Vulnerable User Safety
(10% reduction - based on 8.95× finding)

~NZD 800K

Stream 3: Labour Weekend Enforcement
(15% reduction - Labour Weekend 2014-2017)

05 Top 6 Priority Recommendations

Based on: Gi* hotspot analysis · 8.95× VU relative risk · Labour Weekend finding · SHAP feature importance

CRITICAL

R1



Street Lighting at Hot Spot 99.9% Cells

★ Basis: Lighting = #1 SHAP predictor (V=0.103)

ROI: NZD 2.4M+

CRITICAL

R2



Dedicated Cycling/Pedestrian Infrastructure

★ Basis: Road Lane Type V=0.125 — strongest modifiable factor

ROI: NZD 3.2M+

HIGH

R3



Traffic Signal Upgrades at Top 20 Intersections

★ Basis: Intersection significant (p<0.001); nil control = highest risk

ROI: NZD 1.8M+

HIGH

R4



Labour Weekend Enforcement on State Highways

★ Basis: 6.3% severe rate (+1.47pp above baseline of 4.8%)

ROI: NZD 800K+

MEDIUM

R5



ML Patrol Routing (high_risk_flag cells)

★ Basis: XGBoost recall=63.8%; identifies 6/10 severe crashes

ROI: NZD 1.2M+

MEDIUM

R6



Motorcycle Safety Programme on State Highways

★ Basis: Motorcyclists over-represented on SH; V=0.087

ROI: NZD 1.2M+

06 Ethical & Cultural Safeguards

LO3 - How ethical and cultural considerations were embedded throughout the project

Data Privacy & Security

- ✓ NZTA CAS = no personal identifiers — low base risk
- ✓ Grid-level (1km) aggregation prevents re-identification
- ✓ Cell suppression: $n < 5$ crashes not shown on dashboard
- ✓ Microsoft Fabric encryption at rest + Azure NZ/AU residency

Te Tiriti & Māori Data Sovereignty

- ✓ South Auckland Māori/Pacific communities over-represented in crash data
- ✓ Tino Rangatiratanga: project doesn't speak for affected communities
- ✓ Active Protection: historical underinvestment bias documented
- ✓ R12 monitoring metrics to be co-designed with tangata whenua

Algorithmic Ethics

- ✓ ML predictions = patrol location support only
- ✓ Human officer final decision - not automated enforcement
- ✓ NZ Police Algorithmic Impact Assessment required pre-deploy
- ✓ SHAP explanations ensure model is interpretable, not a black box

Broader Societal Impact

- ✓ CAS reflects crashes that occurred - not suppressed near-misses
- ✓ Under-reporting more common in communities with lower police trust
- ✓ 12 recommendations prioritise infrastructure benefit not surveillance
- ✓ Recommend active travel demand modelling as next-step complement

06 Updated Ethical Risk Register

New risks identified during the research phase - beyond Assessment 2 scope

Risk	Severity	Trigger / Context	Mitigation Applied
Predictive policing misuse	HIGH	ML patrol routing could be misused for demographic targeting in South Auckland	Location-based only; Police Algorithmic Impact Assessment required before deployment; human final decision
Algorithmic bias — South Auckland	HIGH	Model trained on data reflecting historical underinvestment may underestimate risk	Bias explicitly documented in model docs; dashboard limitation card; co-design with community
Data re-identification (rural)	MEDIUM	Rural crashes with unique local context could identify individuals	1km grid aggregation for all public outputs; cell suppression for $n < 5$
Māori data sovereignty	MEDIUM	Crash data disproportionately reflects Māori communities without their governance	R12 monitoring co-designed with tangata whenua; findings don't attribute causes to cultural factors
Dashboard misinterpretation	LOW	Non-technical users may misread AUC, Cramér's V or confidence levels	Plain-language tooltips on all KPI cards; mandatory methodology note cards on slides 3 & 4

07 Conclusion & Next Steps



36 hotspot cells identified - SH1 Southern Motorway corridor confirmed at 99.9% confidence via two independent methods (KDE + Gi*)



XGBoost achieves ROC-AUC 0.718, recall 63.8% - meaningful predictive signal within public data constraints. Lighting and VU involvement = strongest severity predictors



Holiday effect: statistically significant ($p=0.018$) but practically negligible ($V=0.0095$). Labour Weekend (+1.47pp) is the actionable exception



8.95× relative risk for vulnerable users - the strongest single finding. Road lane type and lighting are the strongest modifiable factors



NZD 15–20M estimated annual ROI from 12 evidence-based recommendations across 3 independent intervention streams

Next Steps:

NZTA real-time API integration · Police Algorithmic Impact Assessment for ML deployment · Community co-design of monitoring metrics with tangata whenua · Active travel demand modelling to complement crash data analysis

Key References (APA 7th Edition)

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Q&A Preparation - Likely Questions

Prepared responses demonstrating depth of understanding



Q: Why didn't you achieve the ROC-AUC > 0.85 target from Assessment 1?

A: The public CAS dataset lacks crash hour, driver speed, and alcohol data - the primary severity predictors in literature. All three models converge at ~0.72, confirming a data ceiling, not a model failure. This is consistent with published benchmarks for equivalent feature sets (Mannering et al., 2020).



Q: Isn't using ML for police patrol routing ethically problematic?

A: Yes - this risk was identified and mitigated. The recommendation frames ML as a location-level patrol support tool, not automated enforcement. Human officer final decision is required. A formal NZ Police Algorithmic Impact Assessment must precede deployment. Predictions identify roads, not individuals.



Q: The holiday chi-square is significant but Cramér's V is negligible - which should we act on?

A: Both are correct but measure different things. Statistical significance ($p=0.018$) reflects the large sample size. Cramér's $V=0.0095$ tells us the practical effect is tiny. The actionable conclusion is targeted Labour Weekend enforcement - not calendar-wide resource reallocation.



Q: How did you address Te Tiriti o Waitangi obligations in a technical data project?

A: Three ways: (1) Tino Rangatiratanga - findings don't attribute crash causes to cultural factors. (2) Active Protection - South Auckland bias is explicitly documented. (3) Partnership - R12 recommends co-designing monitoring metrics with tangata whenua representatives, not imposing Crown-defined KPIs.

Thank You

Questions & Discussion

111,657

Crash Records Analysed

36

Hot Spot Cells Identified

8.95×

VU Relative Risk

12

Evidence-Based
Recommendations

NZD 15–20M

Estimated Annual ROI

Sneha Umrikar · Master of Business Informatics · Yoobee Colleges · March 2026

Data-Driven Road Safety Analytics: Predicting and Mapping High-Severity Crash Locations in Auckland

MBI908 Assessment 3 · Final Research Report & Presentation